

Quality of Service (Part 1)

- Topics covered:
 - IP Phones/Voice VLANs
 - Power over Ethernet (POE)
 - Intro to Quality of Service (QOS)
 - Editing ACLs

IP Phones/Voice VLANs

IP Phones?

Traditional phones operate over the **Public Switched Telephone Network (PSTN)**, sometimes called **Plain Old Telephone Service (POTS)**.

IP phones use **Voice over IP (VoIP)** technology to make phone calls over IP networks such as the Internet.

1. Audio data is encapsulated into IP packets.
2. These packets are transmitted across the IP network.
3. Cisco IP phones are commonly used in enterprise networks, although many other vendors also produce IP phones.



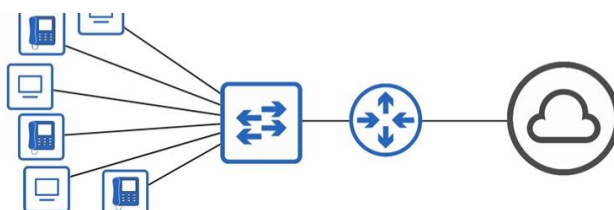
Connecting IP Phones

IP phones connect to a switch just like any other end host.

One possible design is:

- One PC connected to one switch port.
- One IP phone connected to another switch port.

This works, but there is a more common and efficient design.



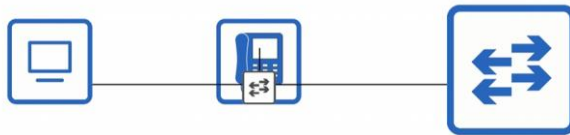
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Internal 3-Port Switch in an IP Phone

An IP phone contains an internal 3-port switch.

It includes:

1. One **uplink** connected to the external switch.
2. One **downlink** connected to the PC.
3. One internal connection to the phone itself.



The internal connection between the switch and the phone is built into the device and does not require an Ethernet cable.

Why This Design Is Better

Because of the internal switch:

- The PC and IP phone share a single switch port.
- PC traffic passes through the IP phone to the switch.
- Phone traffic also reaches the switch through the same connection.



This reduces the number of switch ports required.

Instead of connecting every PC and phone separately, one switch port can support both devices.

Voice VLANs

Voice traffic should be separated from normal data traffic.

- **Voice traffic** = IP phone traffic.
- **Data traffic** = PC traffic.

This separation is done by placing them into different VLANs.

Later, during QoS configuration, voice traffic can be identified and given higher priority than normal data traffic.

VLAN Operation

After configuring a Voice VLAN:

- PC traffic remains **untagged**.
- IP phone traffic is **tagged** with a VLAN ID.

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Configuring a Voice VLAN

Configuration requires only one additional command beyond a normal access port.

Configuration

```
SW1(config)#interface gigabitethernet0/0
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#switchport voice vlan 11
```

PC1 will send traffic untagged, as normal. SW1 will use CDP to tell PH1 to tag PH1's traffic in VLAN 11.

Traffic Flow:

- PC1 sends traffic **untagged**, just like a normal access port.
- SW1 uses **Cisco Discovery Protocol (CDP)** to tell the IP phone to tag its own traffic with **VLAN 11**.



Is This a Trunk Port?

Although the interface carries traffic from:

- VLAN 10
- VLAN 11

it is **not** considered a trunk.

The interface is still configured as an **access port**.

Verifying the Interface

```
SW1#show interfaces g0/0 switchport
Name: Gi0/0
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negoti
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (VLAN0010)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: 11 (VLAN0011)
![output omitted]
```

Although the interface sends/receives traffic from two VLANs, it is not considered a trunk port. It is considered an access port.

Explanation:

1. Administrative Mode is **static access**.
2. Operational Mode is **static access**.
3. Access VLAN is **10**.
4. Voice VLAN is **11**.

Even though the interface carries traffic from two VLANs, it is still considered an **access port**, not a trunk.

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Confirming It Is Not a Trunk

```
SW1#show interfaces trunk
SW1#
SW1#show interfaces g0/0 trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	off	negotiate	not-trunking	1


```
Port      Vlans allowed on trunk
Gi0/0     10-11

Port      Vlans allowed and active in management domain
Gi0/0     10-11

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10-11
```

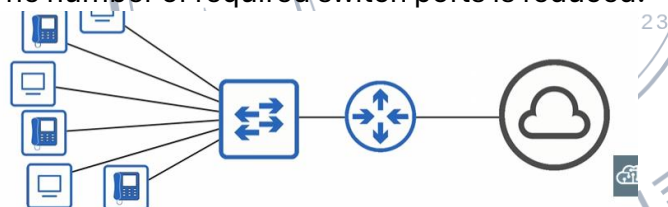
Explanation

1. show interfaces trunk displays nothing, confirming that **G0/0 is not a trunk port**.
2. show interfaces g0/0 trunk shows the interface status as **not-trunking**.
3. Although the output includes "**Vlans allowed on trunk**", this command always displays that field even when the interface is **not** a trunk.

Resulting Network Design

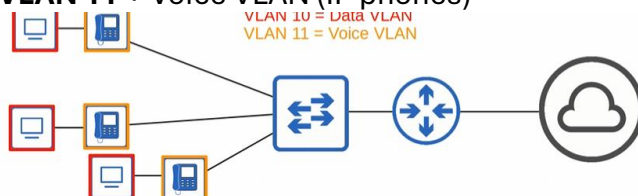
Instead of using one switch port for every PC and another for every phone:

- Each PC connects through its IP phone.
- The number of required switch ports is reduced.



Configuration:

- **VLAN 10** → Data VLAN (PCs)
- **VLAN 11** → Voice VLAN (IP phones)



Power over Ethernet (PoE)

Need of Power over Ethernet?

IP phones require electrical power.

Instead of connecting each phone to a separate power outlet, power can be delivered through the Ethernet cable.

This technology is called **Power over Ethernet (PoE)**.

PoE Components

1. **PSE (Power Sourcing Equipment)** supplies power.
2. **PD (Powered Device)** receives power.

Typically:

- Switch = PSE
- IP Phone = PD

The same Ethernet cable carries:

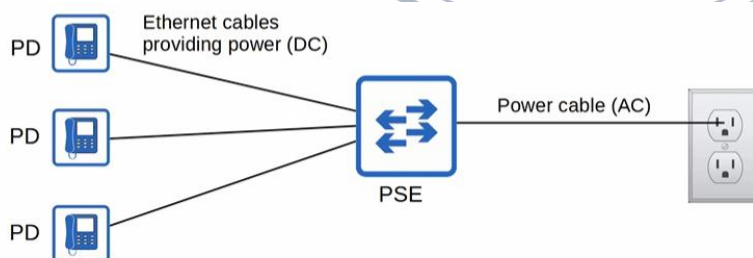
- Data
- Electrical power

Power Flow

The switch receives **AC power** from the wall outlet.

The switch converts: AC Power → DC Power

The resulting DC power is delivered to powered devices over Ethernet.



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Safe Power Delivery

PoE does not immediately provide full electrical power.

Instead, the process is:

1. A device connects to a PoE-enabled port.
2. The PSE sends very low-power signals.
3. The response is monitored.
4. The PSE determines whether power is needed and how much.
5. The PD receives power and boots.
6. The PSE continues monitoring and supplying the required amount of power.

The initial signals are intentionally very weak so connected devices are not damaged.

Power Policing

Power policing prevents a powered device from drawing too much power.

This is one PoE feature to be aware of for the CCNA.

Default Power Policing

Configuration



power inline police configures power policing with the default settings: disable the port and send a Syslog message if a PD draws too much power.

→ equivalent to **power inline police action err-disable**

→ the interface will be put in an 'error-disabled' state and can be re-enabled with **shutdown** followed by **no shutdown**.

CLI Output

```
SW1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)# int g0/0
SW1(config-if)# power inline police
SW1(config-if)# end
SW1# show power inline police g0/0
Available:800(w) Used:32(w) Remaining:768(w)
Interface Admin Oper Admin Oper Cutoff Oper
          State State Police Police Power Power
-----
Gi2/1    auto  on   errdisable ok      17.2  16.7
```

Explanation:

The output displays:

- Available power
- Used power
- Remaining power

Even though action err-disable was not explicitly configured, the default action is err-disable

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Power Policing with Logging

Configuration: **power inline police action log**

If the powered device draws too much power:

- The interface is **not** shut down.
- A **Syslog** message is generated.
- The interface restarts.
- The powered device temporarily loses power and restarts.
- Power negotiation is performed again.

CLI Output

```
SW1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)# int g0/0
SW1(config-if)# power inline police action log
SW1(config-if)# end
SW1# show power inline police g0/0
Available:800(w) Used:32(w) Remaining:768(w)
Interface Admin Oper Admin Oper Cutoff Oper
          State State Police Police Power Power
-----
Gi0/0     auto  on      log      ok      17.2  16.7
```

Explanation

1. Available, used, and remaining power are displayed.
2. The administrative policy is now **log**.
3. If the powered device draws too much power, the interface is **not** placed into the error-disabled state.

For the CCNA, understand the purpose of **power policing**: It prevents powered devices from drawing excessive power from the switch.

PoE Standards

Initially, Power over Ethernet was not known as PoE. Cisco first developed its own proprietary technology called **Cisco Inline Power (ILP)**. Similar to technologies such as **CDP** that were later standardized, Cisco created ILP for its own devices before an industry standard was introduced.

Cisco ILP supplies **7 watts** of power per switch port (PSE) using **2 wire pairs**.

Previously, Ethernet and FastEthernet used only four of the eight wires in an Ethernet cable to transmit data:

- Data transmission: **Wires 1, 2, 3, and 6**
- Power delivery (Cisco ILP): **Wires 4, 5, 7, and 8**

Later, Cisco ILP became the standardized **Power over Ethernet (PoE)** so that other manufacturers could also use the technology.

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PoE Standards

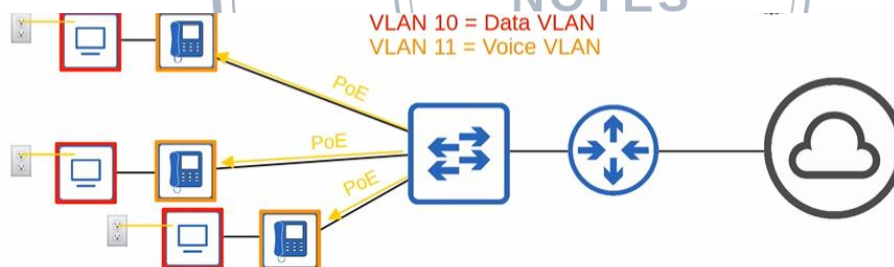
Name	Standard #	Watts	Powered Wire Pairs
Cisco Inline Power (ILP)	Made by Cisco, not standard	7	2
PoE (Type 1)	802.3af	15	2
PoE+ (Type 2)	802.3at	30	2
UPoE (Type 3)	802.3bt	60	4
UPoE+ (Type 4)	802.3bt	100	4

The purpose is to understand the PoE concepts, what PoE is, and what it is used for. Memorizing every value in the table is not emphasized.

PoE in the Network

PCs still require power from a wall outlet.

IP phones, however, receive power through the same Ethernet cable that connects them to the switch.



PoE is not a major CCNA topic, but understanding the concepts presented is expected.

Intro to Quality of Service

Use of Quality of Service?

The discussion of **Voice VLANs** and **PoE** leads into **Quality of Service (QoS)** because QoS is commonly applied to voice traffic from IP phones.

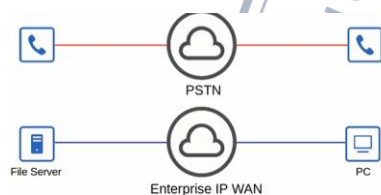
The goal is to give voice traffic a **higher priority** than regular network traffic so that audio quality remains high.

Why QoS Was Created

Originally, voice and data traffic used completely separate networks.

- Voice traffic used the **Public Switched Telephone Network (PSTN)**.
- Data traffic used IP networks such as an **Enterprise WAN** or the **Internet**.

Because the two types of traffic used different networks, they did not compete for bandwidth. As a result, Internet usage did not affect the quality of telephone calls.



Converged Networks

Modern networks are typically **converged networks**.

Voice traffic, video traffic, and regular IP data all share the same IP network.

The PSTN still exists, but the focus here is on the shared IP network.



Converged networks provide:

1. Cost savings.
2. More advanced voice and video features.
3. Integration with collaboration software such as Cisco WebEx and Microsoft Teams.

However, all traffic now competes for the same network bandwidth. If sufficient bandwidth is available and the network is not congested, this is not a problem. During congestion, however, voice and video traffic can be negatively affected because they are sensitive to delay.

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Quality of Service (QoS)

QoS is a collection of tools used by network devices to apply **different treatment to different packets**.

1. Higher-priority traffic receives higher-priority treatment.
2. Lower-priority traffic receives lower-priority treatment.

QoS manages four characteristics of network traffic.

1. Bandwidth

Bandwidth is the total capacity of a network link.

It is measured in **bits per second (bps)**, for example:

- Kilobits per second (Kbps)
- Megabits per second (Mbps)
- Gigabits per second (Gbps)

QoS allows part of a link's bandwidth to be reserved for specific traffic types.

Example allocation:

- Voice traffic → **20%**
- Important data traffic → **30%**
- All remaining traffic → **50%**

2. Delay

Delay is the amount of time required for traffic to travel across the network.

There are two measurements.

One-Way Delay: Measures the time required for traffic to travel from the source to the destination.



The time required for a packet to travel from Phone 1 to Phone 2 is called one-way delay.

Two-Way Delay: Measures the complete round trip.

It includes:

1. Source → Destination
2. Destination → Source (reply)

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Two-way delay measures not only the time taken for the packet to reach the destination, but also the time required for the reply to return to the sender.

3. Jitter

Jitter is the **variation in one-way delay** between packets sent by the same application.

For example:

- Some packets arrive in **10 ms**.
- Other packets arrive in **100 ms**.

A large difference in arrival time means there is **high jitter**.

Effect on Voice Traffic

High jitter negatively affects the quality of phone calls.

To reduce this effect, IP phones use a **jitter buffer**, which provides a fixed delay to incoming audio packets.

If the jitter becomes too high, it overruns the jitter buffer, causing audio quality to suffer.

4. Loss

Loss is the percentage of transmitted packets that never reach their destination.

Packet loss can occur because of:

- Faulty cables.
- Network congestion that causes packet queues to become full.

When a queue becomes full, packets that cannot fit into the queue are discarded.

If audio packets are lost during a phone call, the quality of the call is negatively affected.

Recommended Standards for Interactive Audio

Interactive audio includes communication such as: Phone calls, Zoom calls

Recommended values are:

One-way delay: 150 ms or less
Jitter: 30 ms or less
Loss: 1% or less

When these recommendations are met, users can expect a good experience. If they are not met, the quality of the phone call and the overall user experience may noticeably decrease.

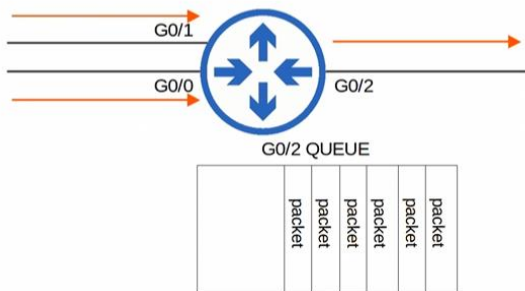
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Queuing

Queuing is an important concept for understanding QoS.

When a network device receives packets **faster than it can forward them**, the packets are temporarily stored in a queue.

For example, a router receives packets on interfaces **G0/0** and **G0/1** faster than it can send them out through G0/2. As a result, the G0/2 output queue begins to fill.



Packets wait in the G0/2 queue before being forwarded.

FIFO (First In, First Out)

By default, packets in the queue are forwarded using First In, First Out (FIFO).

This means:

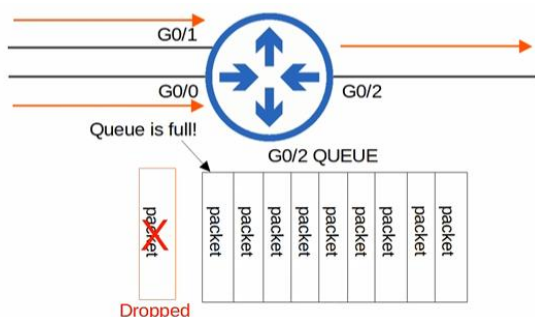
- Packets leave the queue in the same order they arrived.
- No traffic receives special treatment or priority.

Tail Drop

If traffic continues arriving faster than it can be forwarded, the queue eventually becomes full.

When another packet arrives and there is no available space, that packet is discarded.

This behavior is called **tail drop**.



Tail drop occurs because there is no remaining room in the queue for additional packets.

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TCP Global Synchronization

Tail drop causes more than just packet loss.

TCP uses a **sliding window** to control the transmission rate.

The process is:

1. TCP gradually increases its transmission rate.
2. If a packet is dropped, the packet is retransmitted.
3. The sender reduces its transmission rate.
4. The sender gradually increases the rate again.
5. The process repeats whenever another packet is dropped.

Effect of Tail Drop

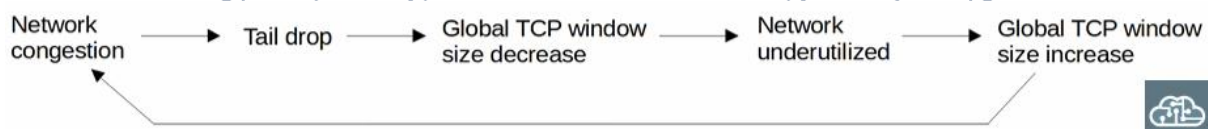
When tail drop occurs because the queue is full:

1. All TCP hosts reduce their transmission rates.
2. Later, they all increase their transmission rates again.
3. This creates repeated cycles of congestion.

As a result, the network alternates between:

- Underutilization
- Congestion

instead of maintaining a stable transmission rate.



Random Early Detection (RED)

A solution to reduce tail drop and prevent TCP global synchronization is **Random Early Detection (RED)**.

Instead of waiting until the queue is completely full:

- The device monitors the queue.
- When the queue reaches a configured threshold, it begins randomly dropping packets from selected TCP flows.

Why RED Is Better?

Only the TCP flows whose packets are dropped reduce their transmission rates.

This prevents **all** TCP flows from reducing and increasing their transmission rates simultaneously, helping keep network traffic more even.

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Weighted Random Early Detection (WRED)

Standard RED treats all traffic equally.

- A single global threshold is used.
- When the threshold is exceeded, packets are randomly dropped regardless of traffic type.

Weighted Random Early Detection (WRED)

Improves upon RED by allowing packet drops to depend on the **traffic class**.

For example:

- HTTP packets can begin dropping at one queue level.
- FTP packets can begin dropping at a different queue level.

Traffic configured as **lower priority** is dropped sooner than higher-priority traffic.

